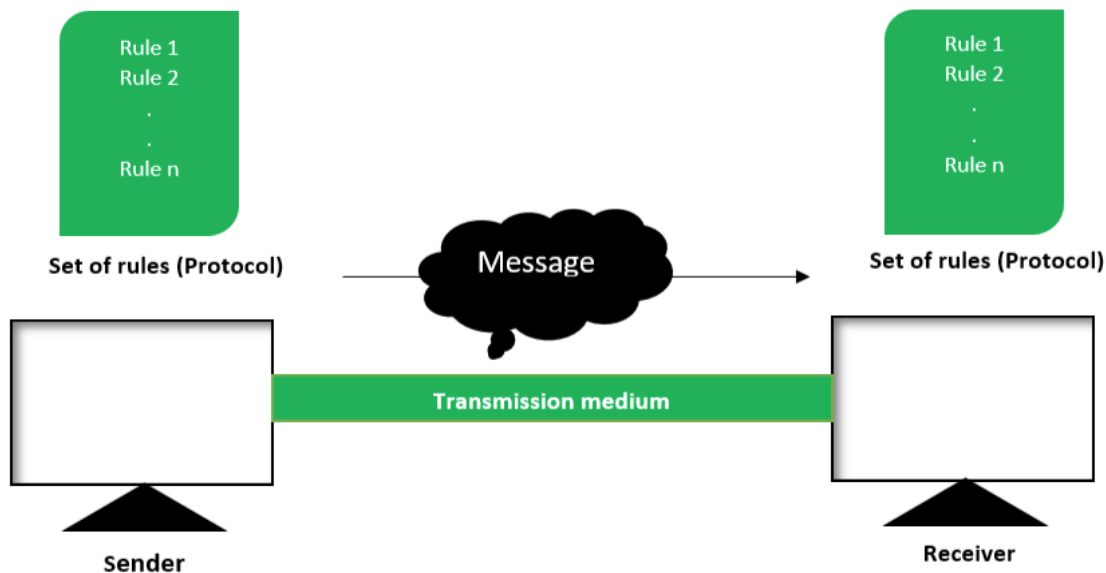


2 Marks

1) Define Data Communication with suitable diagram

Ans. Data communication is a process of exchanging data or information between two devices (a sender and a receiver) through some kind of transmission medium such as co-axial cable or fiber optic cable (wired communication) and/ or air (wireless communication).



2) Enlist four basic characteristics for efficient data communication system.

Ans. 1. Delivery.

The system must deliver data to the correct destination.

2. Accuracy.

The system must deliver the data accurately.

3. Timeliness.

The system must deliver data in a timely manner.

4. Jitter.

Jitter refers to the variation in the packet arrival time. It is the uneven delay in the delivery of audio or video packets.

3) A signal carries three bits in each signal element. If 1200 signal elements are sent per second, find the baud rate and the bit rate.

Ans.

Baud rate = Number of signal elements
= 1200 bps

Bit rate = baud rate * Number of bits per signal element
= 1200 * 3
= 3600 bps

4) Mention any two applications of computer network.

Ans.

- **Resource Sharing:** Networks allow multiple users to access and use hardware and software resources, which reduces costs and improves efficiency.

Examples: Sharing a single printer or scanner across an office, or using centralized storage devices like Network Attached Storage (NAS).

- **Communication:** Networks provide the infrastructure for individuals and organizations to exchange information quickly across any distance.

Examples: Email services (e.g., Gmail), instant messaging (e.g., WhatsApp), and video conferencing (e.g., Zoom).

5) State the difference between baud rate and bit rate.(any 2 points)

Ans.

Feature 	Bit Rate	Baud Rate
Definition	The total number of bits (binary digits) transmitted per unit time.	The total number of signal state changes or symbols transmitted per unit time.
Unit	Bits per second (bps, Kbps, Mbps, etc.).	Baud (Bd) or symbols per second.
Usage	Used to measure the actual data throughput and computer efficiency.	Used to determine the bandwidth required for signal transmission.
Formula	Bit Rate = Baud Rate × Number of bits per symbol.	Baud Rate = Bit Rate / Number of bits per symbol.

4 Marks

1) Compare analog signal and digital signal on any four parameters. (any 4 points)

Ans.

Basis	Analog Signal	Digital Signal
Definition	Analog signals represent continuous variations in magnitude over time.	Digital signals are Discrete and quantized, with specific values.
Signal Type	Continuous waveforms	Discrete Signals
Processing	Requires complex processing for manipulation.	Easier to process and manipulate digitally.
Storage	Less efficient for storage due to continuous nature.	More efficient for storage due to discrete values.
Bandwidth	Typically requires more bandwidth.	Requires less bandwidth for transmission.
Examples	Analog audio signals, analog radio waves, Human voice, etc.	Digital audio signals, digital data streams, computers, etc.

2) Explain the need of computer network. (any 4 points)

Ans.

- 1. Resource Sharing:** Networks allow multiple users to share expensive hardware, such as high-end printers, scanners, and large storage drives (NAS). This significantly reduces equipment costs and physical clutter.
- 2. Effective Communication:** They provide the foundation for real-time communication tools like email, instant messaging, and high-definition video conferencing.
- 3. Centralized Data Management:** Instead of duplicating files across many devices, organizations can store information on central servers. This ensures that every authorized user has access to the most up-to-date version of a document and simplifies data backups.
- 4. Cost-Efficiency:** Beyond hardware savings, networks lower the cost of transferring data. Sending information instantly across the globe via a network is far cheaper and faster than traditional physical methods.

3) Explain protocols and standards used in data communication.

Ans:

1. Protocol:

A protocol is a set of formal rules governing how data is formatted, transmitted, and received.

Syntax: The structure or format of the data (e.g., the first 8 bits might be the sender's address).

Semantics: The meaning of the bits and the actions to be taken (e.g., whether an address is a final destination or just a route).

Timing: When data is sent and how fast it travels to prevent overwhelming the receiver.

Common Protocol Examples:

TCP/IP (Transmission Control Protocol/Internet Protocol): The fundamental suite of the internet. TCP ensures data arrives accurately and in order, while IP handles addressing and routing.

HTTP/HTTPS: Protocols for transferring web pages. HTTPS adds a layer of encryption for security.

2. Standards:

Standards are the universally accepted guidelines that ensure hardware and software from different manufacturers are compatible (interoperable).

Categories of Standards:

De Jure (By Law): Standards officially recognized and legislated by organizations (e.g., ISO, IEEE, IETF).

De Facto (By Fact): Standards that become widely used without official approval, often established by dominant manufacturers (e.g., early versions of specific networking technologies)

Common Standards Organizations:

IEEE: Known for standards like 802.11 (Wi-Fi) and 802.3 (Ethernet).

W3C: Develops standards for the World Wide Web, such as HTML and CSS.

6 Marks

1) A company wants to connect computers within a building, across a city, and across countries. Identify the suitable type of network for each case and justify your answer.

Ans.

Computer network is a collection of interconnected devices that communicate with each other to share resources, data and applications.

Computer network is further classified into three networks

- 1) LAN : within a building
- 2) MAN : across a city
- 3) WAN: across countries

1) Local Area Network (LAN)

A LAN connects computers and devices within a small area like a home, office, school, or hospital. It usually uses switches, routers, and private IP addresses. LANs are high-speed, inexpensive to set up, and easy to maintain.

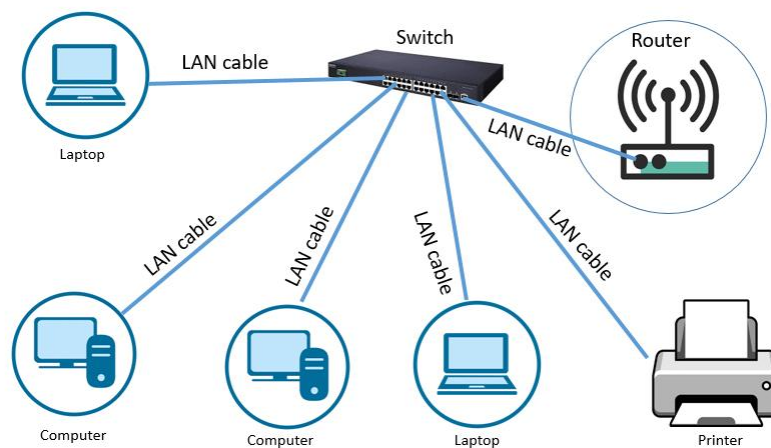
- Coverage: Up to 2 km (limited area).
- Speed: Earlier 4–16 Mbps; now 100–1000 Mbps.
- Medium: Mostly wired (Ethernet cables, twisted-pair, coaxial), though wireless can also be used.

Advantages:

- Provides fast data transfer rates and high-speed communication.
- Easy to set up and manage.
- Can be used to share peripheral devices such as printers and scanners.

Disadvantages:

- Limited geographical coverage.
- Limited scalability and may require significant infrastructure upgrades to accommodate growth.



Local Area Network

2) Metropolitan Area Network (MAN)

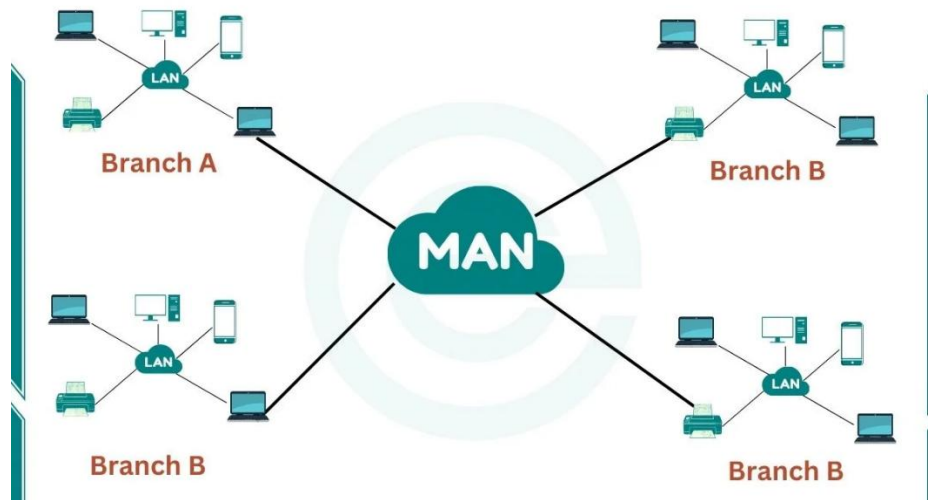
A MAN spans 5–50 km, covering more area than a LAN but less than a WAN. It connects computers across a city or between nearby cities. MANs provide high-speed connectivity (in Mbps), can act as ISPs, and are useful for organizations needing fast communication. However, they are costly, complex to design, and harder to maintain.

Advantages:

- Provides high-speed connectivity over a larger geographical area than LAN.
- Can be used as an ISP for multiple customers.
- Offers higher data transfer rates than WAN in some cases.

Disadvantages:

- Can be expensive to set up and maintain.
- May experience congestion and network performance issues with increased usage.
- May have limited fault tolerance and security compared to LANs.



3) Wide Area Network (WAN)

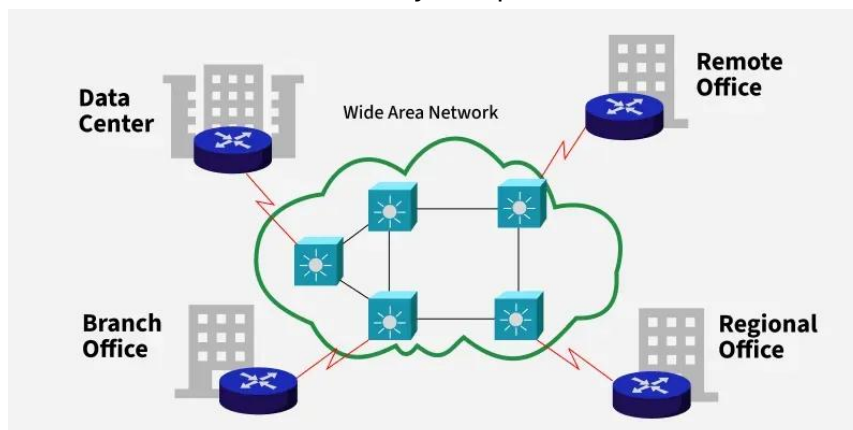
A WAN covers large geographical areas (above 50 km), often connecting multiple LANs through telephone lines, radio waves, or satellites. It can be private (for organizations) or public (like the internet). WANs offer high-speed communication but are costly to set up and maintain.

Advantages:

- Covers large geographical areas and can connect remote locations.
- Provides connectivity to the internet.
- Offers remote access to resources and applications.

Disadvantages:

- Can be expensive to set up and maintain.
- Offers slower data transfer rates than LAN or MAN.
- May have lower fault tolerance and security compared to LANs



2) For the following communication scenarios, identify the mode of communication and justify:

- Online video lecture
- Email communication
- Radio broadcasting

OR

Explain communication / Transmission modes in data transmission

OR

Explain Simplex, Half Duplex and Full Duplex in data transmission.

Ans.

a) Online video lecture: Full Duplex

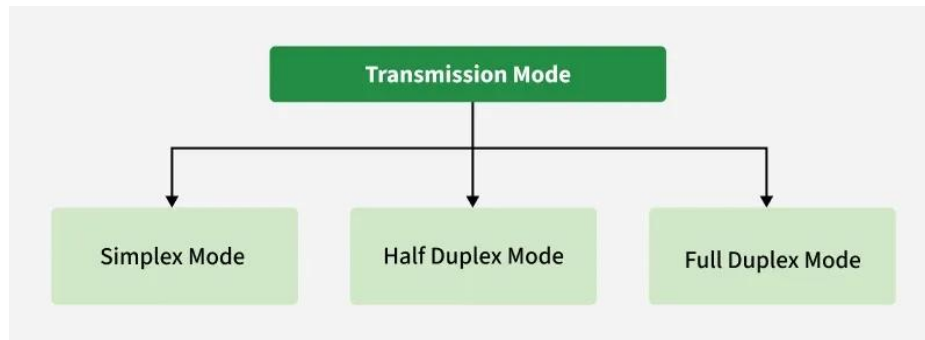
Justification: In an online video lecture, the lecturer speaks and students listen, but students can also ask questions or provide feedback simultaneously, and the lecturer can hear them at the same time they are speaking. This allows for simultaneous, two-way communication.

b) Email communication: Half Duplex

Justification: Email communication allows for two-way communication, but not simultaneously. A person sends a message, and the recipient receives it and then sends a reply. The communication channel is used in one direction at a time, similar to a walkie-talkie conversation where one person speaks, and the other listens before responding.

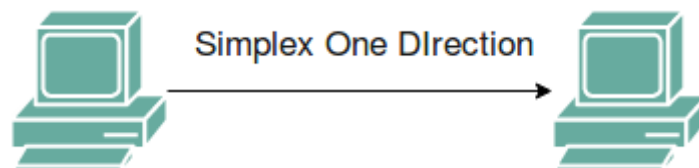
c) Radio broadcasting: Simplex

Justification: In radio broadcasting (e.g., an FM radio station), information is transmitted in only one direction: from the single broadcasting station to a large audience of receivers. The listeners cannot transmit back to the station on the same channel.



Simplex Mode

In Simplex mode, the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit, the other can only receive. The simplex mode can use the entire capacity of the channel to send data in one direction.



Example: Keyboard and traditional monitors. The keyboard can only introduce input, the monitor can only give the output.

Advantages of Simplex Mode

- It is the most cost-effective mode, as it only requires one communication channel.
- There is no need for coordination between the transmitting and receiving devices, which simplifies the communication process.

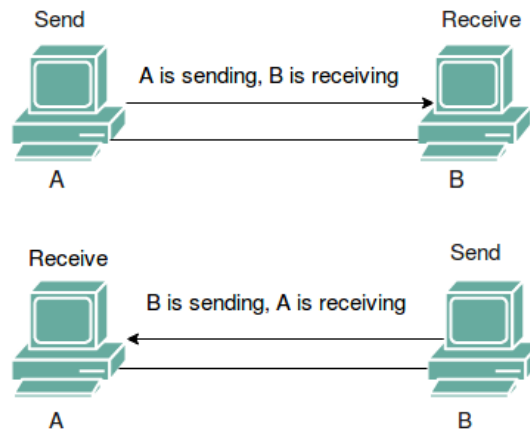
Disadvantage of Simplex Mode

- Only one-way communication is possible.
- There is no way to verify if the transmitted data has been received correctly.
- Simplex mode is not suitable for applications that require bidirectional communication.

Half-Duplex Mode

In half-duplex mode, each station can both transmit and receive, but not at the same time. When one device is sending, the other can only receive, and vice versa. The half-duplex mode is used in cases where there is no need for communication in both directions at the same time. The entire capacity of the channel can be utilized for each direction.

Channel Capacity = Bandwidth* Propagation delay



Example: Walkie-talkie in which message is sent one at a time and messages are sent in both directions.

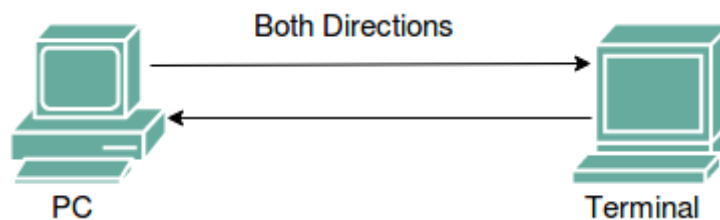
Full-Duplex Mode

In full-duplex mode, both stations can transmit and receive simultaneously. It is used when communication in both directions is required all the time. The capacity of the channel, however, must be divided between the two directions. Here, signals going in one direction share the capacity of the link with signals going in another direction, this sharing can occur in two ways:

Either the link must contain two physically separate transmission paths, one for sending and the other for receiving.

Or the capacity is divided between signals traveling in both directions.

Channel Capacity = $2 * \text{Bandwidth} * \text{propagation Delay}$



Example: Telephone Network in which there is communication between two persons by a telephone line, through which both can talk and listen at the same time.

Advantage of Full-Duplex Mode

- Ideal for real-time applications such as video conferencing or online gaming.
- Provides a high level of reliability and accuracy, as there is no need for error correction mechanisms.

Disadvantage of Full-Duplex Mode

- It's the most expensive mode, as it requires two communication channels.
- It is more complex than simplex and half-duplex modes, as it requires two physically separate transmission paths or a division of channel capacity.
- Full-duplex mode may not be suitable for all applications, as it requires a high level of bandwidth and may not be necessary for some types of communication.

Chapter 2

2 Marks

1) List any two types of guided transmission media.

Ans.

- 1) Twisted Pair cable
- 2) Coaxial cable
- 3) Fiber optic cable

2) Write any two applications of radio waves.

Ans.

Communication and Broadcasting: Radio waves are the fundamental medium for AM and FM radio, as well as television broadcasting.

RADAR and Navigation: **RADAR (Radio Detection and Ranging)** systems use radio waves to detect the position, speed, and distance of objects like aircraft and ships.

3) Define multiplexing.

Ans.

Multiplexing (or muxing) is a communication technique that combines multiple analog or digital signals into a single composite signal for transmission over a shared physical medium.

4) State any two advantages of optical fiber cable.

Ans.

- **Higher Bandwidth and Speed:** Optical fibers provide exceptionally high data-carrying capacity and transmission speeds.
- **Immunity to Electromagnetic Interference (EMI):** Because these cables are made of dielectric materials like glass or plastic and transmit data as light pulses rather than electrical signals, they are completely immune to electromagnetic and radio frequency interference.

5) Difference between Guided and Unguided transmission medium (any 2 points)

Ans.

Guided Media	Unguided Media
The guided media is also called wired communication or bounded transmission media.	The unguided media is also called wireless communication or unbounded transmission media.
The signal energy propagates through wires in guided media.	The signal energy propagates through the air in unguided media.

4 Marks

1) Compare circuit switching and packet switching on the following parameters:

Path establishment

Bandwidth utilization

Delay

Reliability

Parameter	Circuit Switching	Packet Switching
Path establishment	A dedicated communication path is established before data transmission begins	No dedicated path is established; packets may follow different routes
Bandwidth utilization	Inefficient , because bandwidth remains reserved even when no data is transmitted	Efficient , because bandwidth is used only when packets are transmitted
Delay	Low and constant delay after the path is established	Variable delay due to routing, congestion, and queuing
Reliability	Less reliable , as failure of the path breaks the communication	More reliable , as packets can use alternate paths if one path fails

2) Explain the construction of twisted pair cable with neat sketch.

Ans.

- Twisted-pair cables are a type of guided media.
- They have two conductors that are generally made up of copper and each conductor has insulation. These two conductors are twisted together thus, the name twisted pair cables.
- One of the conductors is used to carry the signal and the other is used as a ground reference only.
- The receiver uses the difference of signals between these two conductors. The noise or crosstalk in the two parallel conductors is high but this is greatly reduced in twisted pair cables due to the twisting characteristic.
- In the first twist, one conductor is near to noise source and the other is far from the source but in the next twist the reverse happens and the resultant noise is very less and hence the balance in signal quality is maintained and the receiver receives very less or no noise.
- The quality of signal in twisted pair cables greatly depends upon the number of twists per unit length of the cable.



3) Explain line-of-sight transmission with suitable example.

Ans.

Line-of-sight (LoS) transmission is a communication method where data or signals travel in a straight path directly from the transmitter to the receiver. For this to work effectively, there must be no physical obstructions, such as buildings, mountains, or dense foliage, blocking the "visual" path between the two points.

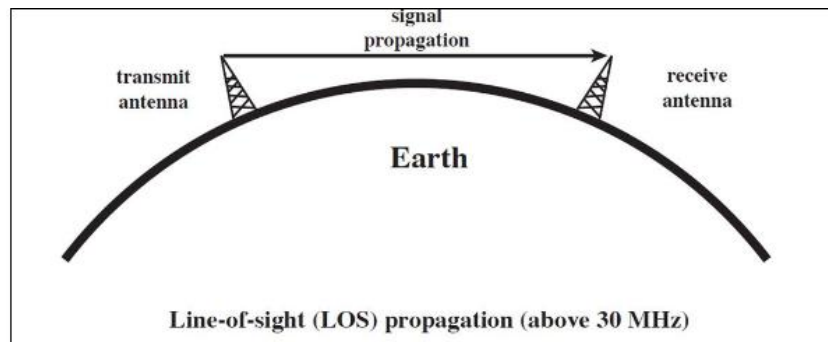
Suitable Example: Television Remote Control

An Infrared (IR) Remote Control is a classic everyday example of line-of-sight transmission.

How it Works: When you press a button, the remote sends a beam of infrared light (invisible to the human eye) directly to the sensor on your TV.

The LoS Requirement: The remote must be pointed toward the TV. If someone stands between you and the TV, or if you try to change the channel from another room, the signal is blocked.

Why it Fails: IR waves have very high frequencies and cannot pass through solid objects like walls or furniture.



6 Marks

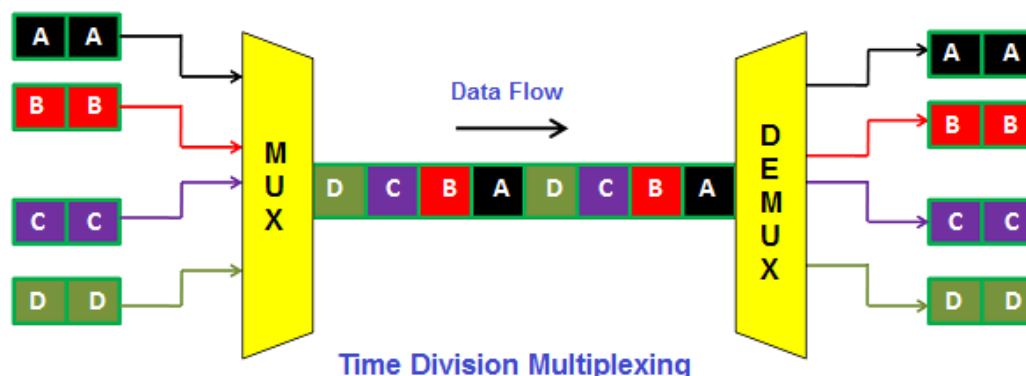
1) Define Time Division Multiplexing (TDM) Explain two types of Time Division Multiplexing with neat sketch. State two applications of TDM.

Ans.

Multiplexing is a technique that allows multiple signals to share a single communication channel simultaneously in order to make efficient use of the available bandwidth.

Time Division Multiplexing (TDM) is one of the techniques of multiplexing.

Time Division Multiplexing (TDM) is a multiplexing technique in which the total available time of a communication channel is divided into different time slots, and each signal is transmitted one at a time using these time slots.



TDM is further divided into two types

1) Synchronous Time Division Multiplexing

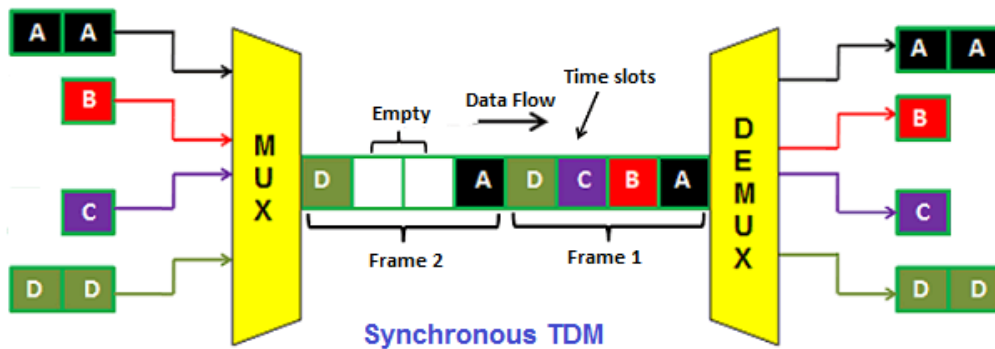
- Time is divided into **fixed and equal time slots**
- Each sender is assigned a **specific time slot**
- Time slots are transmitted in a **fixed order**
- Even if a sender has no data, its time slot is still transmitted
- This may lead to **wastage of bandwidth**

Features:

- Fixed frame structure
- Requires synchronization
- Simple and predictable

Example:

- Digital telephone systems



2) Statistical Time Division Multiplexing (Asynchronous TDM)

Explanation:

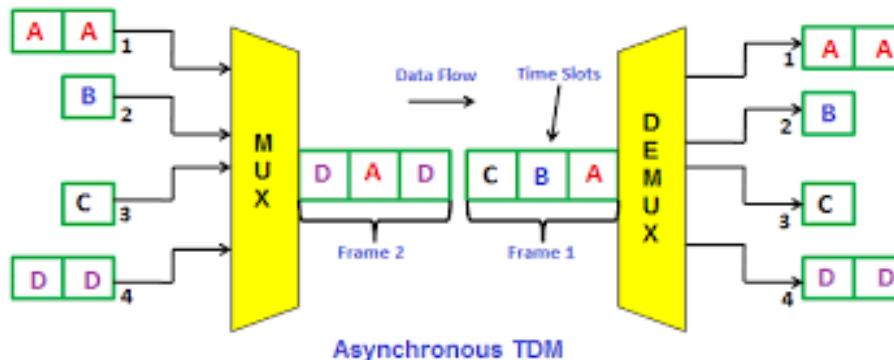
- Time slots are **not fixed**
- Only **active senders** are assigned time slots
- Time slots are allocated **dynamically**
- Each data unit contains an **address** to identify the sender
- Bandwidth utilization is **more efficient** than synchronous TDM

Features:

- No unused time slots
- Better bandwidth utilization
- More complex than synchronous TDM

Example:

- Computer networks and Internet communication



Applications of Time Division Multiplexing (Any Two)

1. Digital telephone networks
2. Computer data communication

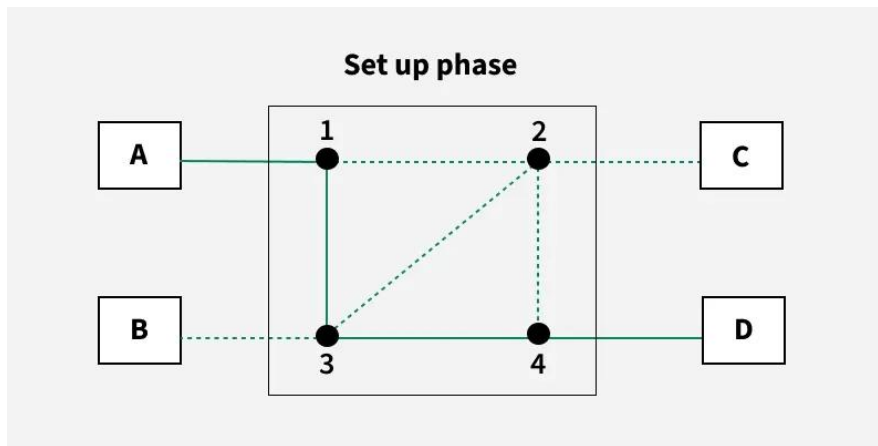
2) Explain three phases of connection oriented packet switching with neat labelled sketch.

Ans.

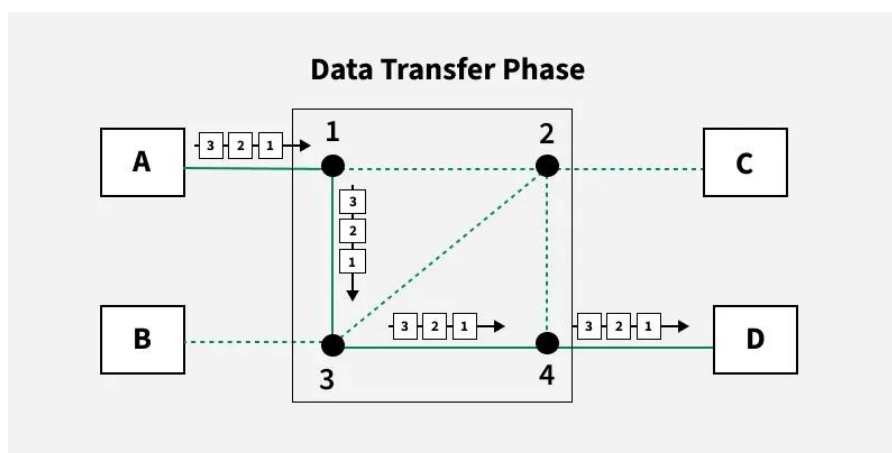
Connection-oriented (Virtual Circuit) Packet Switching establishes a logical path between sender and receiver before data transfer. All packets follow this predefined route and are given sequence numbers to ensure they arrive in order. Each connection is identified by a Virtual Circuit ID assigned by the network.

This method involves three main phases:

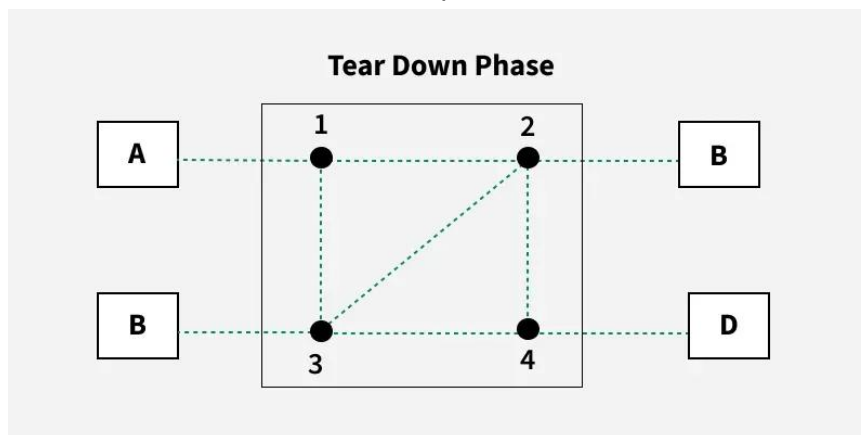
1. Setup Phase: A path is established between sender and receiver. Address information is exchanged and recorded.



2. Data Transfer Phase: Packets are transmitted along the established route, with headers containing local information like length, timestamp, and sequence number.



3. Tear Down Phase: After the transmission is complete, the virtual circuit is released.



Advantages

- **Reliable Data Delivery:** Guarantees packets arrive in order and without loss, crucial for accuracy.
- **Guaranteed Sequencing:** Ensures data packets are reassembled correctly at the destination.

Disadvantages

- **Latency:** The three-phase process (setup, transfer, teardown) adds time compared to direct circuit paths, and queuing delays can occur.
- **Packet Loss/Congestion:** Network overload can lead to packet drops or significant delays, requiring retransmission and impacting quality.